

FL-07 Build Log: Working Agent MVP

PROJECT: ResearchScout Agent | TRACK: General AI Fluency | PHASE: Build (Core) | AUTHOR: Abdul Sami

1. Core Job & Narrow Scope

Narrow Core Job: Given an arXiv paper ID/query, autonomously fetch paper metadata via arXiv REST API, synthesize an engineering research brief, and persist it to local filesystem as structured Markdown.

Live Tools Connected: (1) fetch_arxiv_paper (arXiv REST API + Atom XML parser), (2) save_brief_to_file (Local Filesystem IO).

2. Iteration Log & What Broke

- **Windows Terminal Encoding Error:** Initial console print statements with Unicode status emojis failed with UnicodeEncodeError on Windows CP1252 codepage. Resolved by standardizing terminal logging to clean ASCII tags.
- **arXiv API XML Parsing:** arXiv returns Atom XML format. Implemented namespace handling (<http://www.w3.org/2005/Atom>) to extract title, abstract, authors, and publish date without truncation.
- **Spec Adjustment (Latency Optimization):** Postponed heavy multi-page PDF parsing in favor of raw abstract + structured synthesis template. Reduced execution runtime from 15s to 1.8s for Checkpoint 1 MVP.

3. Unedited End-to-End Execution Trace

```
[ResearchScout Agent] Initialized. Output directory: work/outputs/research_briefs
=====
[AGENT] RUNNING END-TO-END AGENT LOOP: '2312.00752'
=====
[Tool: fetch_arxiv_paper] Querying arXiv API for: '2312.00752'...
[Tool: fetch_arxiv_paper] Successfully retrieved: 'Mamba: Linear-Time Sequence Modeling...' (2312.00752v2)
[Core Job] Synthesizing research brief for arXiv:2312.00752v2...
[Tool: save_brief_to_file] Writing brief to local filesystem:
'work/outputs/research_briefs/arxiv_2312_00752v2_brief.md'...
[Tool: save_brief_to_file] Saved successfully (3314 bytes).
=====
[AGENT] AGENT EXECUTION COMPLETE!
=====
```

4. Evaluation Criteria Verification

- [x] **End-to-End Execution:** Completed full agent loop without mid-run manual edits.
- [x] **Live Tool Connections:** 2 live tools in active use (arXiv REST API + filesystem IO).
- [x] **Documented Deviations:** Documented PDF parsing deferral for latency optimization.
- [x] **Honest Build Log:** Detailed real iteration, encoding fixes, and verification receipts.